

George Joshi

📍 Arlington, Texas

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ML/AI engineer with production experience building semantic search, real-time classification, and data infrastructure systems, plus current applied research in large-scale audio representation learning at UT Arlington. Focused on shipping scalable ML systems from experimentation through deployment.

Education

University of Texas at Arlington

MS, Computer Science

Expected Dec 2027

Tribhuvan University

BS in Computer Engineering

July 2024

Experience

Research Assistant

ACL2 Lab, University of Texas at Arlington

Arlington, Texas Jan

2026 – Present

- Performed **embedding and unsupervised clustering** on a **900K+** sample large-scale acoustic dataset; built segmentation/isolation pipelines (**96–98% F1** in-domain, **68%** zero-shot) scaled to **110K+** files on a multi-GPU cluster.
- Designed a **VQ-based representation learning** framework comparing tokenization strategies for audio signals; found domain-native tokens reconstruct near-ceiling while cross-domain tokens collapse to noise floor – prepared for **ACL Rolling Review (EMNLP)**.
- Evaluated **SOTA self-supervised embedding models (HuBERT, AVES)** across human and non-human audio domains, quantifying transfer performance and domain-gap failure modes.

AI Engineer

DL.surf

Bhaktapur, Kathmandu

Aug 2024 – July 2025

- One of the **first engineers** at the company; owned backend, ML, and infrastructure systems end-to-end with broad autonomy over technical decisions.
- Built a semantic **Search & Re-ranking system** (**150K+** daily **FastAPI** requests) using **all-MiniLM-L6-v2**, **Milvus**, and **Cohere Reranker**; reduced query latency $\sim$ **60%** via Redis caching.
- Developed a **low-latency NSFW classifier** for images, video, and text integrated with upload/commenting APIs; **quantized models to INT8 (PyTorch → ONNX → TFLite)** to exploit CPU-native SIMD acceleration via **XNNPACK**, avoiding GPU infrastructure costs while meeting latency targets.
- Designed **relational database schemas from scratch** for new platform features and services (**PostgreSQL**); configured **RBAC**, connection pooling, indexing, and stored procedures for reliability at scale.
- Independently **dockerized** all services across the stack and integrated **CI/CD** pipelines (**Docker**, **GitHub Actions**, **Jenkins**), connecting internal services with inter-service dependencies; owned deployment end-to-end.
- Designed and built **public-facing, application-level REST APIs** consumed directly by the product.
- Built **Grafana dashboards** for real-time monitoring: one with **Prometheus + Grafana** for system performance, and another with **Postgres + Grafana** for platform **user metrics**.

Machine Learning Intern

Treeleaf Technologies

Kathmandu, Nepal

May 2023 – Jul 2023

- Developed and optimized **YOLOv7**-based computer vision models for object detection; collaborated on team deployment of ML models to production.
- Built automated pipelines for deploying custom ML models, using **Git**-based version control.

Projects

AniSpeakArchive – Bioacoustics Literature Search Engine

- Built a **4-stage, resumable data pipeline** (ingest → classify → annotate → index) that filtered **100K+** OpenAlex papers down to a relevant bioacoustics corpus using a fine-tuned **SciBERT** classifier run via multi-GPU streaming inference.
- Used **Qwen2.5-14B** (served with **vLLM**) for structured metadata extraction, and built a **hybrid BM25 + dense-embedding** search index (**LanceDB**) with reranking to power the search app.
- Deployed a **Flask** web app with OTP-based auth and saved-search alerts; containerized with **Docker** and served via **Gunicorn** WSGI in production.

Technologies

Languages: Python, SQL, C++, JavaScript

ML/DL: PyTorch, TensorFlow, HuggingFace, scikit-learn, Keras, OpenCV, YOLO, MLflow

Infra & Cloud: AWS (EC2, ECR, S3), Docker, Kubernetes, GitHub Actions, Prometheus, Grafana

Data: PostgreSQL, Milvus, Kafka, MongoDB